

# CEED SOLUTION 2014

Sai Praveen

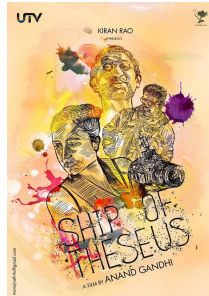
## CEED 2014 PART A

There are 50 questions. Each question carries 2 marks for correct answer,

- 0.5 for incorrect answer.

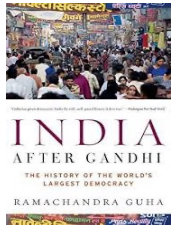
Q1: The film 'Ship of Theseus' was directed by

- A. Anand Patwardhan
- B. Satyajit Ray
- C. Karan Johar
- ☒ D. Anand Gandhi



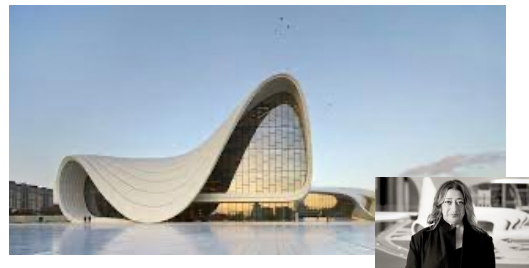
Q2: Author of the book 'India After Gandhi' is

- A. V. S. Ramachandran
- B. Ramachandra Guha
- C. Jawaharlal Nehru
- D. M.J. Akbar



Q3: Zaha Hadid is well known as\_\_\_\_\_.

- A. a fashion photographer
- B. a fashion designer
- C. a singer
- ☒ D. an architect



Q4: The famous temple that was affected in June-July 2013 floods in Uttarakhand is

- A. Badrinath
- B. Somnath
- ☒ C. Kedarnath
- D. Bhimashanker

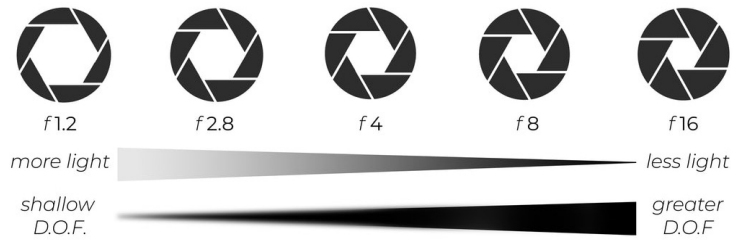
Q5: \_\_\_\_\_ is an example of monolithic temple architecture.

- A. Trimbakeshwar temple
- B. Meenakshi temple
- ☒ C. Kailashnath temple
- D. Akshardham temple



Q6: With a 50 mm lens when you change aperture from 2.8 to 8, the depth of field of the image will

- ☒ A. Increase
- ☐ B. Decrease
- ☐ C. Remain unchanged
- ☐ D. None of the above



Q7: Postscript in digital printing was developed by \_\_\_\_\_

- ☐ A. Apple
- ☐ B. Microsoft
- ☐ C. Google
- ☒ D. Adobe



a short remark or message added to the bottom of a letter after you have signed your name

### Lens Aperture Chart for Beginners

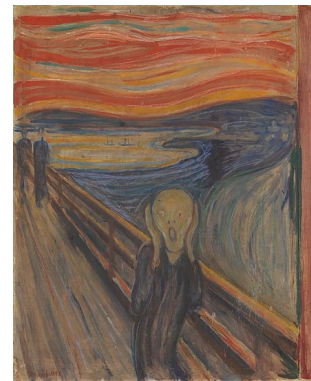
| f/1.4                     | f/2.8                | f/5.6                 | f/11                 | f/22                      |
|---------------------------|----------------------|-----------------------|----------------------|---------------------------|
| Very Large Aperture       | Large Aperture       | Medium Aperture       | Small Aperture       | Very Small Aperture       |
| Very Small Depth of Field | Small Depth of Field | Medium Depth of Field | Large Depth of Field | Very Large Depth of Field |
| Almost Nothing In Focus   | Little In Focus      | Some In Focus         | Much In Focus        | Almost All In Focus       |
|                           |                      |                       |                      |                           |
| Brightest                 | Bright               | Medium                | Dark                 | Darkest                   |

Q8: The following lithograph was done by \_\_\_\_\_.

**Lithography is a planographic method of printing originally based on the immiscibility of oil and water. The printing is from a stone or a metal plate with a smooth surface. It was invented in 1796 by the German author and actor Alois Senefelder and was initially used mostly for musical scores and maps.**



- ☐ A. Kathe Kollwitz
- ☒ B. Edward Munch
- ☐ C. Yeng Li Pen
- ☐ D. Wu Tao Zu



Q9: Which of the following animals is not depicted In Ashoka Pillar ?

- ☐ A. Bull
- ☐ B. Horse
- ☐ C. Lion
- ☒ D. Tiger

animals demonstrated on the pillar are horse, bull, elephant, and lion.



Q 10: Identify the 'India Design Mark' symbol.



A



B



C



☒ D

Q11: Identify the famous Indian artist who painted the picture below



- A. Ghulam Mohammad Shaikh
- B. Amrita Shergil
- ☒ C. Raja Ravi Varma
- D. Abhinindranath Tagore



Q 12: Identify the following monument



- A. Bharhut Stupa
- ☒ B. Sanchi Stupa
- C. Kushinagar Stupa
- D. Bavikond Stupa

Q13. One of the methods of 3D printing is called \_\_\_\_\_

- A. Laser Microscopy
- B. Surface scanning microscopy
- ☒ C. Stereolithography
- D. Offset Printing

Stereolithography (SLA or SL; also known as vat photopolymerisation, optical fabrication, photo-solidification, or resin printing) is a form of 3D printing technology used for creating models, prototypes, patterns, and production parts in a layer by layer fashion using photochemical processes by which light causes chemical monomers and oligomers to cross-link together to form polymers. Those polymers then make up the body of a three-dimensional solid. Research in the area had been conducted during the 1970s, but the term was coined by Chuck Hull in 1984 when he applied for a patent on the process, which was granted in 1986. Stereolithography can be used to create prototypes for products in development, medical models, and computer hardware, as well as in many other applications. While stereolithography is fast and can produce almost any design, it can be expensive.

Q14. Boot space of a car refers to \_\_\_\_\_

- ☒ A. Luggage space
- B. Engine Envelope
- C. Place to keep shoes
- D. Petrol Tank



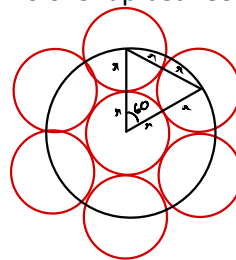
Q15. Identify the organization which uses the symbol shown here.



- ☒ A. Trade Fair Authority of India
- B. Indian Institute of Foreign Trade
- C. Indian Institute of Packaging
- D. Trade Union Federation of India

Q16. What is the maximum number of circular discs of equal size, which can be placed around a similar disc, such that the central disc is touched by all others but there is no overlap between any of the discs?

- A. 6
- ☒ B. 8
- C. 12
- D. 10



$$\frac{360}{60} = 6 \text{ Circles}$$

Q17. A platonic solid with 12 sides is called \_\_\_\_\_.

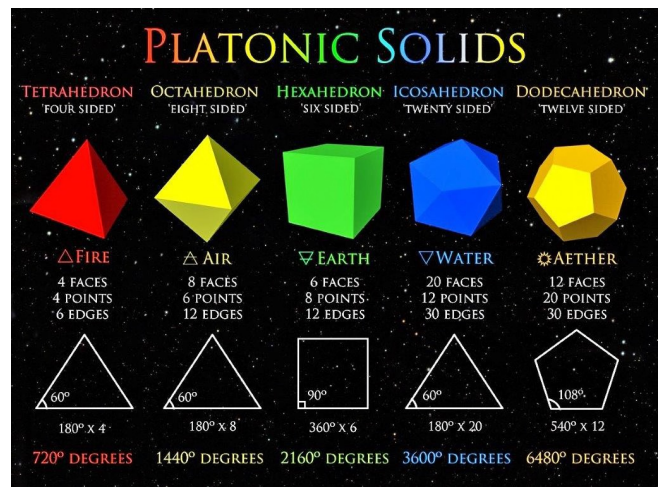
- A. Tetrahedron
- B. Cube
- C. Icosahedron
- ☒ D. Dodecahedron

Q18. Which among these is not a dance form in India?

- A. Kathak
- B. Mohiniyattam
- C. Odissi
- ☒ D. Thumri

Q19. Identify the correct sentence

- A. Which of your two colleagues are here
- ☒ B. Which of your two colleagues is here
- C. Which of two colleagues are here
- D. Which of the two colleagues are here

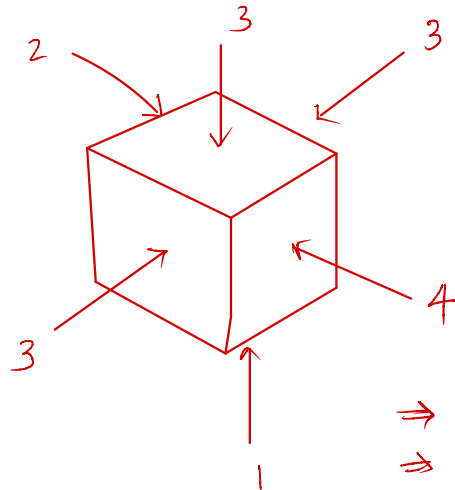
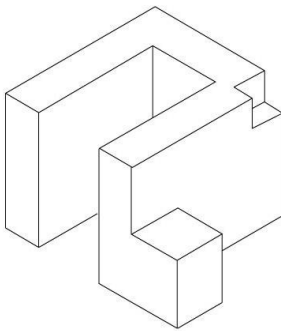


Q20. Complete the sentence:

..... his carelessness our car was ruined

- A. Cause of
- B. Since
- C. For
- ☒ D. Due to

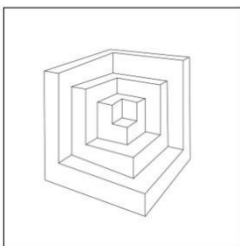
Q21. How many surfaces are there in the illustration shown?



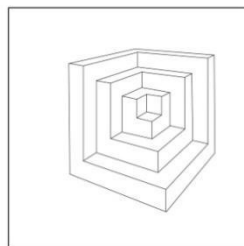
$$\Rightarrow 3 + 3 + 4 + 1 + 3 + 2$$
$$\Rightarrow 16$$

- A. 14
- ☒ B. 16
- C. 17
- D. 15

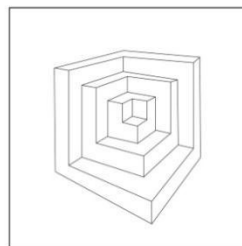
Q22. Identify the drawing with correct perspective.



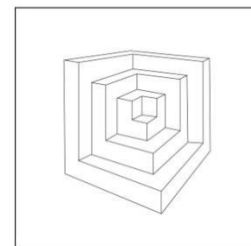
A



B



C



☒ D

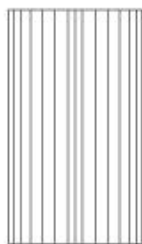
Q23. Identify this tribal art form of India



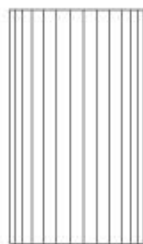
- A. Warli
- B. Saura
- C. Madhubani
- ☒ D. Gond

Q24. Plan of a three dimensional object is shown here.

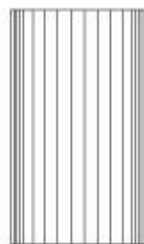
Identify the correct front view of the object from amongst the four images shown below .



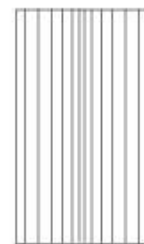
A



☒ B



C



D



Q25. What is the meaning of 'latent'?

- A. Worthy
- ✓ B. Veiled
- C. Latterly
- D. Flanking

GIF

Graphics Interchange Format

JPG

Joint Photographic Group

PNG

Portable Network Graphics

PDF

Portable Document Format

PSD

Photoshop Document

XML

Extensible Markup Language

CSV

Comma-Separated values

TTF

True Type Font

MPEG

Moving Picture Experts Group

Q26. What is the full form of MPEG?

- ✓ A. Motion Pictures Experts Group
- B. Magnetic Prism Exposure Grid
- C. Motion Picture Entertainment Guild
- D. Media Protocol Entertainment Guild

Q27. The film that Eisenstein is associated with \_\_\_\_\_.

- A. Blow up
- B. Bicycle Thieves
- ✓ C. Battleship Potemkin
- D. Roshomon

He is noted in particular for his silent films Strike (1925), Battleship Potemkin (1925) and October (1928), as well as the historical epics Alexander Nevsky (1938) and Ivan the Terrible (1944, 1958).

Q28. Identify the famous graphic designer from the list

- A. Sabyasachi Mukherjee → *fashion designer*
- B. Sudarshan Dheer → *Graphic designer*
- C. Dilip Chhabria → *Automobil designer*
- D. Hafeez Contractor → *Architect*

Q29. Identify the car manufacturer based on the visual shown here



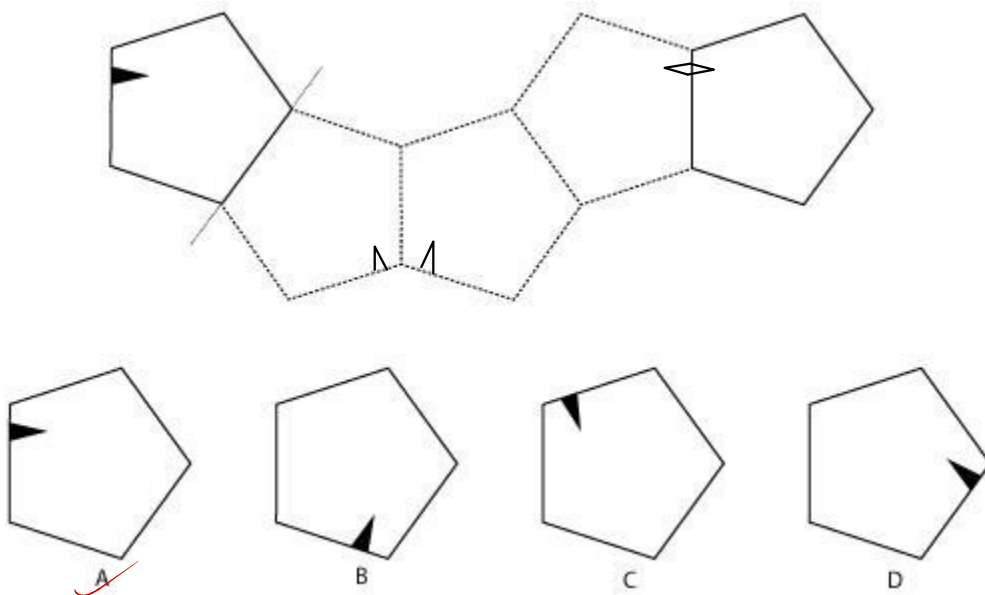
- A. Rolls Royce
- ~~B. BMW~~
- C. Maruti Suzuki
- D. Land Rover

Q30. Identify the car brand from the visual shown

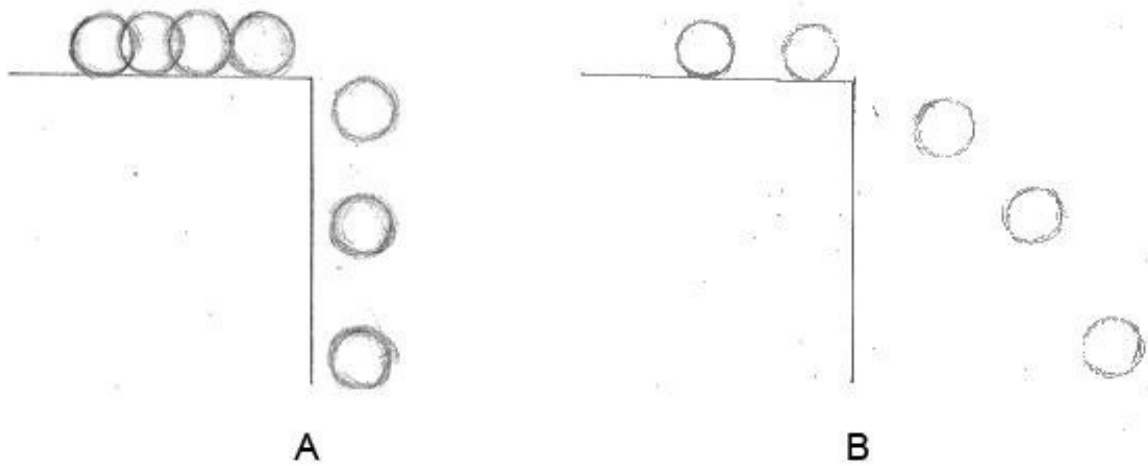


- ~~A. Cadillac~~
- B. Vauxhall
- C. Citroen
- D. Dacia

Q31. A Pentagon is mirrored four times. What will the final image seen in the last Pentagon be?

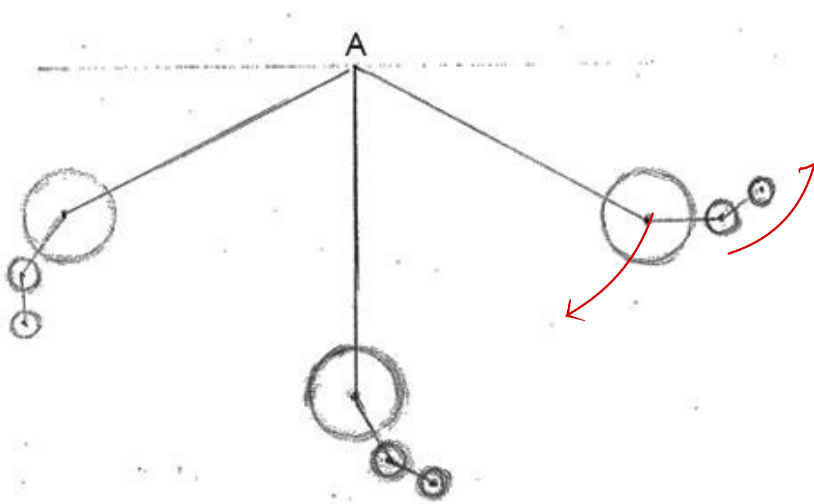


Q32. You are shown two animation sequences (A and B) of a ball falling off a surface. Assuming that they were pushed with the same force, identify which animation sequence depicts the heavier ball ?



- ☒ A. Ball in image A is heavier
- ☐ B. Ball in the image B is heavier
- ☐ C. Ball in image A and B are of equal weight
- ☐ D. Can't say

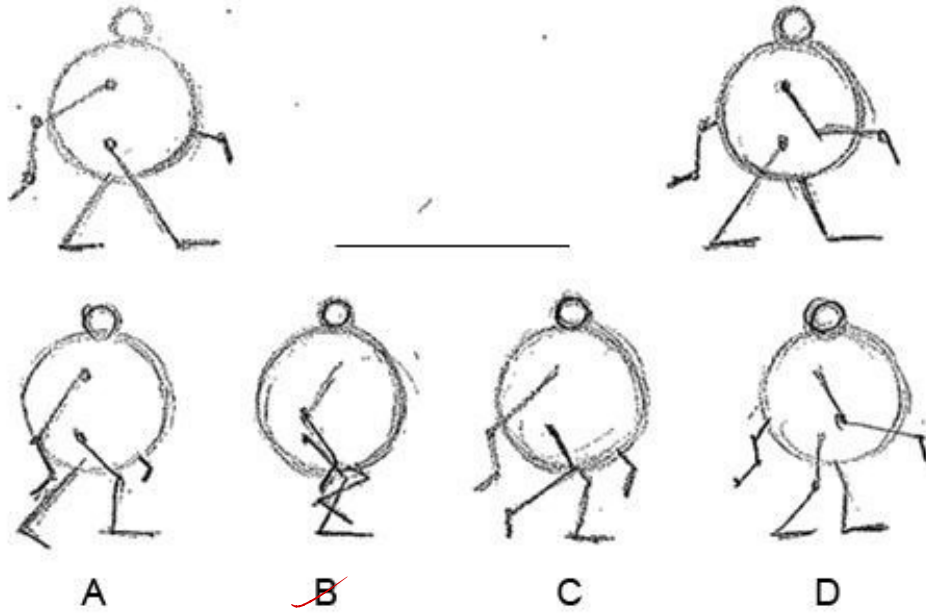
Q33. In the image shown identify the direction of the swinging pendulum around pivot point A.



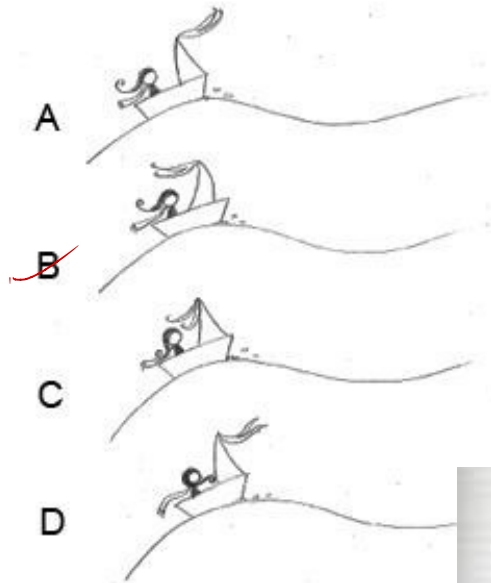
- ☐ A. Left to Right
- ☐ B. Both the directions

- C. None of the above
- ~~D. Right to Left~~

Q34. Choose the correct mid-point image to complete the walk cycle

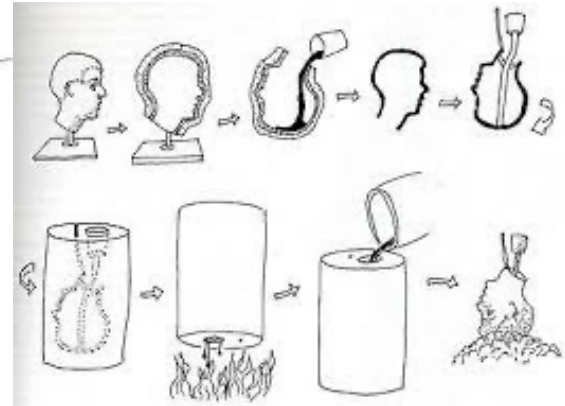


Q35. A girl in a boat is riding a wave. Identify the correct drawing



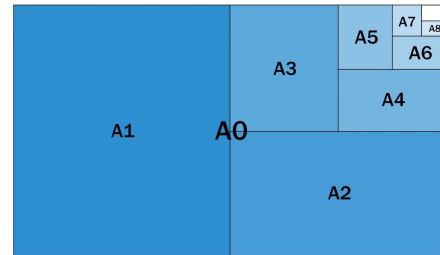
Q36. Lost wax process is used in \_\_\_\_\_.

- A. Turning
- ☒ B. Casting
- C. Forging
- D. Welding



Q37. How many A4 size paper sheets will make A0 size?

- A. 3
- B. 4
- ☒ C. 16
- D. 8



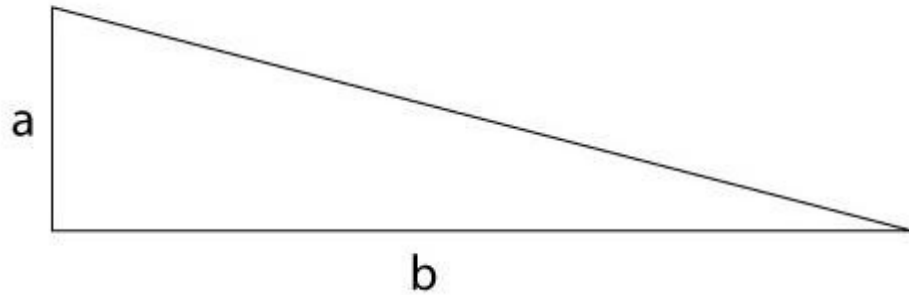
Q38. As per DIN standards, the dimension of A4 size paper is \_\_\_\_\_.

- ☒ A. 210 by 297 mm
- B. 297 by 268 mm
- C. 236 by 278 mm
- D. 210 by 297 inches

| Paper Size | Millimetres   | Centremetres   | Inches        |
|------------|---------------|----------------|---------------|
| A0         | 841 x 1189mm  | 84.1 x 118.9cm | 33.1 x 46.8in |
| A1         | 594 x 841mm   | 59.4 x 84.1cm  | 23.4 x 33.1in |
| A2         | 420 x 594mm   | 42 x 59.4cm    | 16.5 x 23.4in |
| A3         | 297 x 420mm   | 29.7 x 42cm    | 11.7 x 16.5in |
| A4         | 210 x 297mm   | 21 x 29.7cm    | 8.3 x 11.7in  |
| A5         | 148.5 x 210mm | 14.85 x 21cm   | 5.8 x 8.3in   |
| A6         | 105 x 148.5   | 10.5 x 14.85cm | 4.1 x 5.8in   |
| A7         | 74 x 105mm    | 7.4 x 10.5cm   | 2.9 x 4.1in   |
| A8         | 52 x 74mm     | 5.2 x 7.4cm    | 2.0 x 2.9in   |
| A9         | 37 x 52mm     | 3.7 x 5.2cm    | 1.5 x 2.0in   |
| A10        | 26 x 37mm     | 2.6 x 3.7cm    | 1.0 x 1.5in   |



Q39. Maximum acceptable slope (a:b) for wheel chair ramp is in the ratio of \_\_\_\_\_.



(Image not to scale)

- A. 1:8
- ☒ B. 1:12
- C. 1:3
- D. 1:20

**The Americans with Disabilities Act (ADA) requires a 1:12 slope ratio for commercial ramps (ramps used in public spaces). This means that for every 1 inch of rise, 12 inches of ramp are required. For example, if an entryway is 36 inches high, you will need a ramp that is at least 36 feet long. The 1:12 slope ratio reduces the steepness of a ramp to 5 degrees of incline and makes it both easier and safer to traverse.**

**While the ADA requirements only apply to commercial ramps, many home access professionals adhere to the 1:12 slope ratio for residential ramps as well. However, a slope ratio of 2:12 is also acceptable for residential ramps if space is limited. This means 12-inches of ramp for every 2-inches of rise and 9.5 degrees of incline.**

Q40. What is the material used commonly in the manufacture of household plastic buckets?

- A. ABS
- ☒ B. PVC
- C. Nylon
- D. HDPE

Q41. Jiggering & Jollying process is associated with \_\_\_\_\_.

- ☒ A. Ceramics
- B. Mild steel
- C. Aluminum
- D. Bakelite

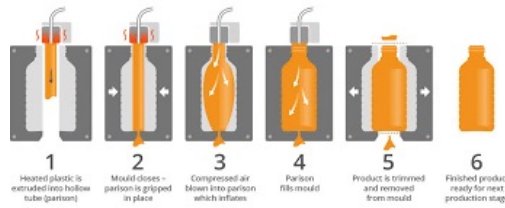
Q42. Carbon Foot Print refers to \_\_\_\_\_.

- ☒ A. Environment Pollution
- B. Photosynthesis in Plants
- C. Carbon dating technique
- D. Carbon Fiber Footwear



Q43. Pet bottles are best made using \_\_\_\_\_ process.

- A. Vacuum Forming
- B. Injection molding
- ☒ C. Blow molding
- D. Roto-molding



Q44. Tyeb Mehta is a well known \_\_\_\_\_.

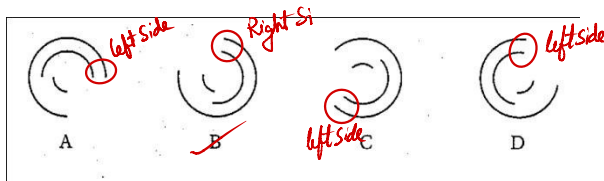
- A. Animator
- ☒ B. Painter
- C. Automobile Designer
- D. Fashion Designer

Q45. The symbol shown is associated with



- A. Amul
- B. Life Insurance Corporation of India
- C. Food Corporation of India
- ☒ D. Nabard

Q46. Identify the odd one out from the list



Q47. The symbol shown is associated with



- A. National Savings Scheme
- B. National Textile Corporation
- ☒ C. Aadhar
- D. National Literacy mission

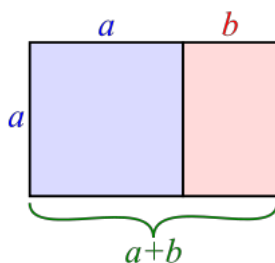
Q48. Identify the musical instrument shown in this picture



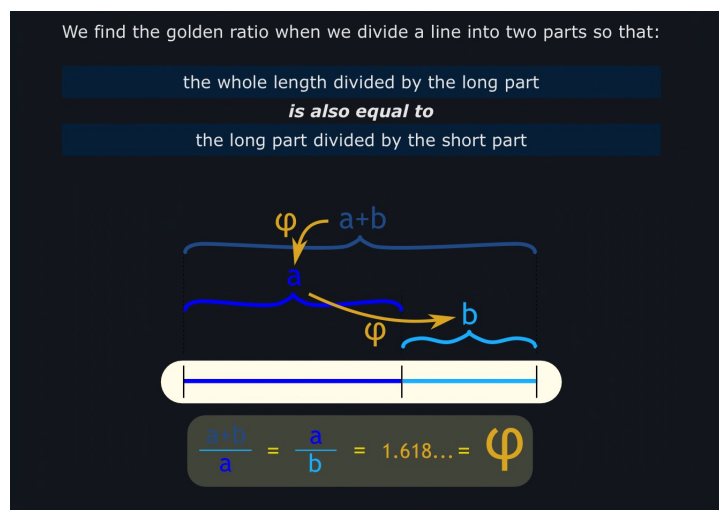
Ustad Amjad Ali Khan (born 9 October 1945) is an Indian classical sarod player, best known for his clear and fast ekhara taans. Khan was born into a classical musical family and has performed internationally since the 1960s. He was awarded India's second highest civilian honor Padma Vibhushan in 2001, India's third highest civilian honor Padma Bhushan in 1991 and Padma Shree in 1975.

- A. Sarangi
- B. Guitar
- C. Sarod
- D. Sitar

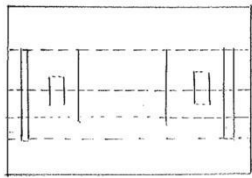
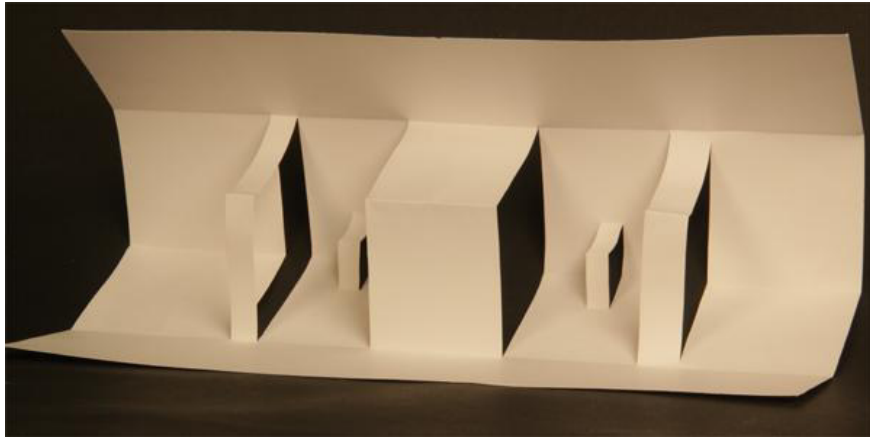
Q49. The visual shown here represents the golden proportion. Identify the correct ratio.



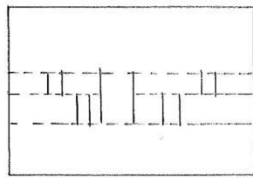
- ☒ A.  $a:a+b = 1: 1.618$
- B.  $a:a+b = 1: 6.182$
- C.  $a:a+b = 1: 4.352$
- D.  $a:a+b = 1: 1.928$



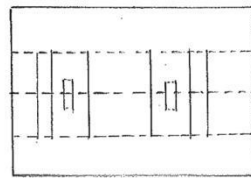
Q50. Identify the correct cut and fold drawing to create the shown image.



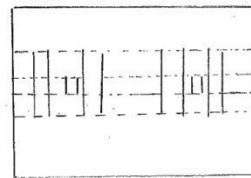
A



B



~~C~~



D

## **CEED 2014 PART B**

**QUESTION NO: 1 TO 3 ARE MANDATORY.**

**ATTEMPT ANY ONE FROM QUESTION NO: 4, 5 AND 6.**

### **Sketching**

**Question No. 1**

**Marks 20**

Draw a **freehand** perspective drawing of a person opening the refrigerator door in the kitchen.

**OR**

Draw a **freehand** perspective drawing of a person standing next to a scooter parked on the street.

**Note:**

- DO NOT USE ANY DRAWING INSTRUMENTS LIKE RULER, COMPASS, SET-SQUARE etc.
- DO NOT USE ANY FORM OF COLOUR
- USE ONLY PENCILS

**Evaluation Criteria:**

- Abilities in perspective drawing
- Light and shade and shadow
- Quality of lines
- Sense of scale and proportions
- Sense of composition

## Creativity and Imagination

### Question 2

Marks 15



The illustration above is an artist's impression of an incident in a club house. Narrate the incident in your own words, based on your creative imagination.

Limit your description to less than 250 words.

#### Evaluation Criteria:

- Originality of the narrative
- Imagination and Ideation
- Story telling ability



### Question 3

Marks 15

Umbrellas are used mainly during the rainy season and usually remain unused during the rest of the year.

Creatively generate 3 alternative uses of the umbrella other than just as protection from rain or sunshine. Illustrate your ideas in the form of sketches with short descriptions.

#### Evaluation Criteria:

- Originality of the ideas
- Creativity and Innovation
- Imagination and Ideation

## **CEED 2014 Part B Optional (Visual Communication and Interaction)**

### **Question 4**

**Marks 50**

**Design a graphical user interface (GUI) for a website of a movie theatre complex named 'Navrang Chitrapat' for booking movie tickets online. The complex has 3 theatres with each theatre screening a different film, three shows per day.**

**Present your design solutions in the form of screen shots. Draw minimum of six screen shots explaining the booking procedure in a sequence.**

**Support your design solution with a brief note on the design decisions.**

**The web interface should present the following information to the user:**

- **Basic Information about the theatre complex**
- **Name of film**
- **Date**
- **Time**
- **Availability of ticket**
- **Price of tickets**
- **Payment Options**
- **Confirmation of Booking**

**Evaluation Criteria:**

- **Usability of the Interface**
- **Structure and clarity of Information**
- **Visual quality of icons and overall screens**
- **Logic of design decisions**

## **CEED 2014 Part B Optional (Animation Design)**

### **Question 5**

**Marks 50**

A woman is walking on a crowded railway platform. A train pulls in and people alight. A man suddenly comes up behind the woman, snatches her bag and runs away. The woman shouts for help.

A policeman standing near a vendor's cart looks up and immediately gives chase.

Two women standing by, try to console the shocked woman. The thief runs towards the stairs to the overhead bridge. The policeman smartly throws his baton at the thief's legs. The thief stumbles and comes crashing down on the platform. A group of men pounce on the thief and catch him. They retrieve the bag as the cop reaches and pins the thief down.

A loud cheer goes up amongst the watching public.

Assume you are one of the characters in this story. Visually illustrate the story as you see it unfolding in front of you through a sequence of minimum 20 frames.

Tell this story from the point of view of any one of the following characters:

- The woman or
- The thief or
- The policeman or
- A bystander standing on the platform

### **Evaluation criteria**

- Visual narrative structure
- Quality of illustrations
- Character and Environment development

## **CEED 2014 Part B Optional (Product Design)**

### **Question 6**

**Marks 50**

**Present road side waste containers are normally dirty, stinking and spilling over. Design roadside waste container bin(s) to be used in public places for segregation of the following categories of waste.**

- **Different types of packaging**
- **Wet and Dry organic waste**
- **Organic food waste**
- **All other kinds of leftover.**

**The container(s) should be rugged, movable so that the waste can be transported for periodic disposal.**

**Present your design proposal**

- With a neat freehand sketch showing the form.**
- An exploded view drawing**
- Indicate the materials and processes used in manufacturing.**
- Support your design with a brief note on the design decisions taken.**

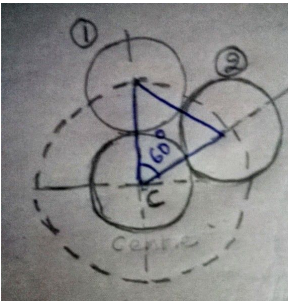
**Evaluation criteria**

- **Identification of problem areas**
- **Design insights and creativity**
- **User Centric Design considerations**
- **Sustainability parameters recycle, reduce, regenerate, refuse**
- **Ability to meet specific requirements of hygiene, portability and Indian conditions**
- **Originality of ideas**
- **Appropriateness of materials to be used in the manufacturing of the final product.**

**CEED 2014**

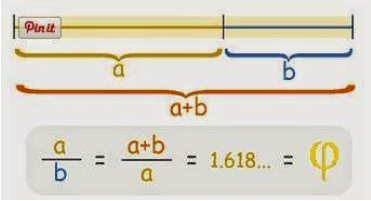
| Q.No | Ans | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Ans image | Learn about more                                                                                                                                                                                    |
|------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | D   | The film 'Ship of Theseus' was directed by 'Anand Gandhi'. The film received positive reviews from both Indian and International press and has been given "The most significant film to come out of India in a very long time" and also got the title "The hidden gem of the year". The film also won the award of 'Best Feature Film' of the year at the National Film Awards. The film received an overall 10 awards and was also nominated for 5 film awards. |           |                                                                                                                                                                                                     |
| 2    | B   | The book 'India after Gandhi' was written by Ramachandra Guha. The book is all about the history of India; after it gained independence from the British. The Book received the title 'Book of the Year' from The Economist, The Wall Street Journal and Outlook. The book also won the 2011 'Sahitya Akademi Award'.                                                                                                                                            |           |                                                                                                                                                                                                     |
| 3    | D   | Zaha Mhd Hadid is an Architect famous for building concepts that are futuristic with more concern on powerful curving forms of the elongated structures with 'Multiple perspective' points. She is the first women to receive Pritzker Architecture prize. She also own the Stirling prizes (twice).<br>Her official website - <a href="#">zaha hadid blog</a>                                                                                                   |           | <a href="http://stuffyoulook.blogspot.com/2014/06/culture-and-architecture-guide-for-ceed-exam.html">http://stuffyoulook.blogspot.com/2014/06/culture-and-architecture-guide-for-ceed-exam.html</a> |
| 4    | C   | On 16th and 17th June 2013, The Kedarnath temple was hit with flash floods, resulting in landslides and mudslides with loud peals of thunder. Near about 70000 pilgrims stranded.                                                                                                                                                                                                                                                                                |           |                                                                                                                                                                                                     |
| 5    | C   | The Kailsahnath temple is a famous <b>monolithic monument</b> (also called as <b>rock-cut</b> ) - one f the 34 monasteries and temples collectively known as the ' <u>Ellora Caves</u> '. Monolithic buildings are buildings/shrines that are carved, cast or excavated from a single piece of material. The most famous being those cut from rocks.<br><br>Learn more about - <a href="#">Famous Rock cut, monolithic structures in India</a>                   |           |                                                                                                                                                                                                     |
| 6    | A   | Depth of field (DOF) can be described in two ways. Shallow and Deep DOF. Shallow is when the included focus range is very narrow. Deep is when the included range is large. f-stops work a inverse values, such that a small f/number (say f/2.8) corresponds to a larger or wider aperture value, which results in a shallow depth of field. A large f/number (say f/16) results in a smaller or narrower aperture size and therefore a deeper                  |           | <a href="http://stuffyoulook.blogspot.com/2013/10/photography-for-CEE-D-Part-A.html">http://stuffyoulook.blogspot.com/2013/10/photography-for-CEE-D-Part-A.html</a>                                 |

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|----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |   | depth of field.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |                                                                                                                                                                                                       |
| 7  | D | PostScript is a computer language for creating Vector graphics. It was developed by the designers of Adobe. PostScript went beyond the typical printer control language and was a complete programming language of its own.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |                                                                                                                                                                                                       |
| 8  | B |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |                                                                                                                                                                                                       |
| 9  | D | <p>Ashoka pillar bear the images of an elephant, a galloping horse, a bull, and a lion. All these animals are separated by overriding wheels of chariot over a bell-shaped lotus. The pillar is carved out of a single block of polished sandstone (monolithic architecture).</p> <p>These animals symbolize the four different phases of Gautama Buddha's life. The Elephant stands for the Buddha's notion in reference to the dream of Queen Maya, a white elephant entering her tomb. The Bull points out the birth of Buddha which happened during the full moon in the month of Vesak under the zodiacal sign Taurus. The Horse denotes the horse Kanthaka, which Buddha used for going away from palatial life. The Lion shows the attainment of enlightenment, the Buddhahood.</p> |  |                                                                                                                                                                                                       |
| 10 | D |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |                                                                                                                                                                                                       |
| 11 | C | The blend of colors, focus on Indian women, culture, and devotional surroundings; itself confirms that those type of paints were the creations of Raja Ravi Varma.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  | <a href="http://stuffyoullook.blogspot.com/2013/10/indian-state-paintings-and-painters.html">http://stuffyoullook.blogspot.com/2013/10/indian-state-paintings-and-painters.html</a>                   |
| 12 | B |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |                                                                                                                                                                                                       |
| 13 | C |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  | <a href="http://stuffyoullook.blogspot.in/2013/10/guide-for-Part-A-resources-GK.html">http://stuffyoullook.blogspot.in/2013/10/guide-for-Part-A-resources-GK.html</a>                                 |
| 14 | A |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |                                                                                                                                                                                                       |
| 15 | A |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  | <a href="http://stuffyoullook.blogspot.com/2014/06/logs-of-Indian-institutes-and-corporations-1.html">http://stuffyoullook.blogspot.com/2014/06/logs-of-Indian-institutes-and-corporations-1.html</a> |

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|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-----------------------------|--------------------------------|-------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--|
| 16                                                                                 | A                                                                                   | Let us just consider two circles (numbered 1 and 2) touching the center disc C. As you can see from the figure, the two circles makes an angle 60 degree with the center. Since, the lines joining the centers are of same length (twice the radius), the lines form an equilateral triangle (check the picture). So, the total degrees (360) with that made with the combination of two adjacent circles (60 degrees) to get 6 as the answer. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                             |                                   |                             |                                |                               |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                     |  |
| 17                                                                                 | D                                                                                   | A platonic solid is a regular polyhedron with same faces of meeting at each vertex.<br><br>The following picture illustrates the five common solids that meet the criteria of Platonic solids.                                                                                                                                                                                                                                                 | <table border="1"> <tr> <td>Tetrahedron<br/>(four faces)</td> <td>Cube or hexahedron<br/>(six faces)</td> <td>Octahedron<br/>(eight faces)</td> <td>Dodecahedron<br/>(twelve faces)</td> <td>Icosahedron<br/>(twenty faces)</td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table> | Tetrahedron<br>(four faces)                                                                                                                                                 | Cube or hexahedron<br>(six faces) | Octahedron<br>(eight faces) | Dodecahedron<br>(twelve faces) | Icosahedron<br>(twenty faces) |  |  |  |  |  |  |
| Tetrahedron<br>(four faces)                                                        | Cube or hexahedron<br>(six faces)                                                   | Octahedron<br>(eight faces)                                                                                                                                                                                                                                                                                                                                                                                                                    | Dodecahedron<br>(twelve faces)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Icosahedron<br>(twenty faces)                                                                                                                                               |                                   |                             |                                |                               |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                     |  |
|  |  |                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                          |                                   |                             |                                |                               |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                     |  |
| 18                                                                                 | D                                                                                   | While the rest (Kathak, Mohiniyattam and oddissi) are Indian classical dances belonging to different parts of India, Tumri is a semi-classical Indian music.                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                             |                                   |                             |                                |                               |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                     |  |
| 19                                                                                 | B                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                             |                                   |                             |                                |                               |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                     |  |
| 20                                                                                 | D                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                             |                                   |                             |                                |                               |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                     |  |
| 21                                                                                 | B                                                                                   | Front side - 3 faces<br>Right side - 4 faces<br>Back side - 3 faces<br>Left side - 2 faces<br>Top - 3 faces<br>bottom - 1 face<br>Total - 16 faces                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <a href="http://stuffyoulook.blogspot.com/2013/09/solving-number-of-faces-on-solid.html">http://stuffyoulook.blogspot.com/2013/09/solving-number-of-faces-on-solid.html</a> |                                   |                             |                                |                               |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                     |  |
| 22                                                                                 | D                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                             |                                   |                             |                                |                               |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                     |  |
| 23                                                                                 | D                                                                                   | Madhubani does not belong to the category of Tribal art.                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <a href="#">Indian state paintings and painters</a>                                                                                                                         |                                   |                             |                                |                               |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                     |  |
| 24                                                                                 | B                                                                                   | First thing first, there won't be any center line for the front view if you had carefully observed the plan section. The width between the two consecutive gear tooth goes on decreasing from center to either side. For further clarification, just count the number of gear tooth on the plan view and equate that to that available in the front view (take twice by considering the bottom also)                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                             |                                   |                             |                                |                               |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                     |  |
| 25                                                                                 | B                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                             |                                   |                             |                                |                               |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                     |  |
| 26                                                                                 | A                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                             |                                   |                             |                                |                               |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                     |  |
| 27                                                                                 | C                                                                                   | The film Battleship Potemkin was directed by Sergei Eisenstein. Eisenstein is a pioneer in the theory and practice of montage. He is also famous for silent films and Battleship Potemkin is one among them.                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                             |                                   |                             |                                |                               |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                     |  |
| 28                                                                                 | B                                                                                   | Sabyasachi Mukherjee is a noted Indian fashion designer. Dillip Chhabria is a Transportation designer. Hafeez Contractor is an Indian Architect.                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                             |                                   |                             |                                |                               |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                     |  |



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|----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------------------|
| 29 | B | Get the logos of Indian automobiles -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  | <a href="#">Indian car manufacturer logos list</a>  |
| 30 | A |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                                                     |
| 31 | A | See the working of similar problem here                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  | <a href="#">working with geometric objects</a>      |
| 32 | A |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                                                     |
| 33 | D | The bigger ball tends to push the medium ball towards the direction of motion and the medium in turn tries to drag the smaller ball. In the picture, it can be seen that the bigger ball is always to the left of the remaining balls and hence obviously, the balls are moving from right to left.                                                                                                                                                                                                                                                                                                                                                                                                                 |  |                                                     |
| 34 | B | The first picture (left one) of the question depicts that the right leg of the image (the leg that is in front to us) is in front and straight position, likewise the left leg (that is on the other side of the person) is straight and is back. So, the person is about to take a step with his right leg. Also note the position of hands. The second image is quite opposite to the first image. Now the middle picture should be in transition between the two posters (legs and hands). The right leg should be in bent position and in fact it should be in middle, touching the ground and the left leg should be in air (he just took step) and probably in bent position. So, position in B option suits. |  | <a href="#">Step-to-step animations</a>             |
| 35 | B |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                                                     |
| 36 | B | Lost wax method also called as 'Investment casting' is a casting process that is used to cast (or make molds) of complex shapes - shapes that can't be made using normal castings because of more shapes/details.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |                                                     |
| 37 | C | The size of paper sheets just goes on doubling.<br>Like<br>A4 paper - 1 sheet<br>A3 paper - 2 A4 sheets (2 A4 sheets)<br>A2 paper - 2 A3 sheets (4 A4 sheets)<br>A1 paper - 2 A2 sheets (8 A4 sheets)<br>A0 paper - 2 A1 sheets (16 A4 sheets)                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |                                                     |
| 38 | A |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                                                     |
| 39 | B |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                                                     |
| 40 | B | Learn more about -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  | <a href="#">Materials and manufacturing process</a> |
| 41 | A |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                                                     |
| 42 | A | A carbon footprint is the total sets of greenhouse gas emissions caused by an organization, event,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |                                                     |

|    |   |                                                                                                                                                                                                                                                                                     |                                                                                     |                                                                           |
|----|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
|    |   | product or person. So, it's related to environmental process/pollution.                                                                                                                                                                                                             |                                                                                     |                                                                           |
| 43 | C | Learn more about -                                                                                                                                                                                                                                                                  |                                                                                     | <a href="#">Materials and manufacturing process</a>                       |
| 44 | B | Learn more about -                                                                                                                                                                                                                                                                  |                                                                                     | <a href="#">Indian state paintings and famous painters</a>                |
| 45 | D | You may go through -                                                                                                                                                                                                                                                                |                                                                                     | <a href="#">Complete Logo list of Indian corporations and institutes</a>  |
| 46 | B | The small arcs in the options (A,B,C) are following medium sized arcs in clockwise direction, only in option C, small arc is following medium sized arc in counter-clockwise direction.                                                                                             |                                                                                     |                                                                           |
| 47 | C | You may go through -                                                                                                                                                                                                                                                                |                                                                                     | <a href="#">Part - II Logo list of Indian corporations and institutes</a> |
| 48 | C | Learn more about -                                                                                                                                                                                                                                                                  |                                                                                     | <a href="#">Indian musical instruments</a>                                |
| 49 | A | <p>The Golden ration "phi" is a special number which is approximately equal to 1.618 and is used in mathematics, physics and architecture. It is the ratio of longer part (total) divided by the smaller part and is equal to 1.618.</p> <p>The below picture shows the details</p> |  |                                                                           |
| 50 | C |                                                                                                                                                                                                                                                                                     |                                                                                     |                                                                           |